
Adversarial Action Removal in Self-Play Reinforcement Learning

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Abstract

We study adversarial action masking in self-play reinforcement learning: an attacker selectively removes actions from a learning agent’s action space to maximise performance degradation. In Kuhn and Leduc Poker, across tabular Q-Learning, PPO, NFSP, and neural DQN victims, we show that: (i) adversarial masking causes up to $9\times$ more damage than random masking at the same budget, and $11\times$ more than RARL-style action perturbation; (ii) the adversary learns to minimise reach-weighted contingent action capacity (CAC_w), empirically confirmed by a Pearson correlation of $r = 0.80$ between CAC_w and victim reward; (iii) self-play co-adaptation amplifies the attack; and (iv) the learned mask transfers across agents, indicating structural vulnerability. Critically, a neural adversary (MLP with REINFORCE) against a DQN victim in Leduc Poker achieves -2.57 reward per hand ($2.2\times$ worse than random masking)—demonstrating the phenomenon scales beyond tabular settings to function-approximation agents in games with ~ 50 information sets.

1 Introduction

Multi-agent reinforcement learning (MARL) agents trained via self-play achieve strong performance in competitive domains [Silver et al., 2018, Brown and Sandholm, 2019], but their robustness to structural environment changes remains poorly understood. Prior work on adversarial attacks focuses on observation perturbations [Huang et al., 2017, Gleave et al., 2020] or reward manipulation [Zhang et al., 2020]. We study a different, more severe attack surface: the *action space* itself.

An adversary that selectively removes actions—disabling specific capabilities rather than adding noise—poses a qualitatively different threat from bounded perturbation. Such attacks arise naturally: hardware failures disable actuators, regulatory changes restrict strategies, API deprecations remove endpoints, and sandboxing constrains agent capabilities. We formalise this as a bi-level optimisation where the inner loop trains an RL agent under masked actions and the outer loop trains an adversary to choose which actions to remove.

Our key finding is that adversarial masking is *dramatically more efficient* than random removal: a neural adversary targeting fewer than 50% of decision points in Leduc Poker causes $9\times$ more damage than removing a fixed action everywhere. The mechanism operates through selective minimisation of reach-weighted contingent action capacity (CAC_w)—empirically verified by strong correlation between CAC_w and victim reward across budget levels.

Contributions.

- We formalise the adversarial action masking problem and show it is fundamentally more severe than action perturbation ($11\times$ more damaging than RARL).

- We demonstrate the attack scales from tabular (Q-Learning, PPO, NFSP) to neural (DQN) agents across games of increasing complexity (Kuhn: 6 states; Leduc: ~ 50 states).
- We provide empirical evidence that the adversary minimises CAC_w (Pearson $r = 0.80$), connecting the attack to game-theoretic structure.
- We show the attack transfers across agents and is amplified by self-play co-adaptation.

2 Related Work

Adversarial attacks on RL. Adversarial observation perturbations [Huang et al., 2017, Gleave et al., 2020] and reward poisoning [Zhang et al., 2020] are well-studied threat models. Action-space attacks are less explored: invalid action masking [Huang and Onta  n, 2022] addresses the opposite problem (preventing illegal actions), and constrained MDPs [Altman, 1999] formalise action restrictions but not adversarial selection. We study *deliberate, strategic* action removal by a learned attacker.

Action-robust and adversarial RL. Robust Adversarial RL (RARL) [Pinto et al., 2017] trains an adversary that *perturbs* the agent’s actions within a bounded set; NR/PR-MDP formulations [Tessler et al., 2019] extend this to probabilistic and nominal-robust action channels. These methods model continuous perturbations where the chosen action is *modified*. Our setting is structurally different: actions are *removed* entirely (discrete, binary availability), which eliminates decision points rather than distorting them. This distinction matters because removal can reduce contingent action capacity to zero—a qualitative regime change that bounded perturbation cannot induce. Observation-adversarial methods [Zhang et al., 2021, Sun et al., 2022] operate on a separate channel.

Self-play robustness and NFSP. Self-play agents can cycle, overfit, or exhibit non-transitive behaviour [Balduzzi et al., 2019, Lanctot et al., 2019]. Population-based methods (PSRO) [Lanctot et al., 2017] maintain diversity to mitigate these. NFSP [Heinrich and Silver, 2016] combines RL with average-strategy tracking. We show that *all* of these—including NFSP—collapse under structural action removal, because no algorithm can compensate when the action space itself is reduced.

3 Problem Formulation

Setup. Consider a two-player zero-sum game with self-play training. Player 0 (victim) learns via RL. An adversary $\mathcal{M}$ produces a mask per decision point:

$$\text{mask} = \mathcal{M}(\text{info_state}, A_{\text{legal}}, \text{player})$$

returning a subset of legal actions. The victim observes only the masked action set and is unaware of the adversary.

Bi-level optimisation.

$$\text{Inner: } \pi^* = \arg \max_{\pi} \mathbb{E} \left[\sum_t r_t \mid \pi, \mathcal{M} \right] \quad (1)$$

$$\text{Outer: } \mathcal{M}^* = \arg \min_{\mathcal{M}} \mathbb{E} \left[\sum_t r_t \mid \pi^*, \mathcal{M} \right] \quad (2)$$

The inner loop trains the victim under the mask; the outer loop updates the adversary to minimise the victim’s expected reward.

Budget constraint. The adversary can mask at most k information sets: $|\{h \in \mathcal{I}_0 : |\mathcal{M}(h)| < |A(h)|\}| \leq k$. This models realistic constraints where an attacker controls limited endpoints.

Connection to CAC_w . Each masked state where $|A| > 1$ is reduced to $|A| = 1$ decreases CAC by 1. The adversary’s objective is equivalent to *greedy minimisation of reach-weighted CAC_w* :

$$\mathcal{M}_k^* = \arg \min_{|\text{supp}(\mathcal{M})| \leq k} CAC_w(\pi_0 \mid \mathcal{M})$$

where $CAC_w = \sum_h \rho(h) \cdot \mathbf{1}[|\mathcal{M}(h)| > 1]$ and $\rho(h)$ is the reach probability of information set h .

Deterministic Exploitation Attractor (DEA). When $CAC_w = 0$, the game reduces to a single-agent MDP with deterministic transitions.

Proposition 1 (DEA Convergence). *Under $CAC_w = 0$ in self-play Q-learning with ε -greedy ($\varepsilon > 0$, $\alpha \in (0, 1)$): (i) the victim’s policy converges to the unique forced action at every information set; (ii) the opponent’s Q-values converge to $Q^* = V^{BR(\sigma^*)}$; (iii) $(\sigma^*, BR(\sigma^*))$ is a stable fixed point.*

Proof sketch. Forced actions make the victim’s policy Q-value-independent. The opponent faces a stationary MDP; Q-learning converges under standard conditions. The pair is stable: neither player can unilaterally deviate. In Kuhn, $BR(\sigma_{\text{forced-pass}}^*) = -1.0$ per hand.

4 Methods

Self-play protocol. Both players share a single learning agent; the mask is applied to Player 0’s actions at every decision point during *both training and evaluation*. In the self-play regime, both sides adapt continuously. In the fixed-opponent regime, a snapshot of the agent’s value function is frozen before masking begins; this static copy plays as Player 1 while only Player 0 continues learning under the mask.

Victim agents. We evaluate four algorithms spanning tabular and neural methods:

- **Tabular Q-Learning:** ε -greedy, $\varepsilon = 0.15$, $\alpha = 0.1$.
- **Tabular PPO:** softmax policy, clipped surrogate, entropy bonus (Appendix A).
- **Tabular NFSP** [Heinrich and Silver, 2016]: best-response QL + average strategy, $\eta = 0.1$.
- **DQN:** 2-layer MLP (64 hidden), experience replay (20k buffer), target network, Adam (10^{-3}). Full hyperparameters in Appendix C.

Table 1: Masking strategies evaluated.

Strategy	Description
None	Full action space (baseline)
Random(p)	Remove each action independently with prob. p
Fixed	Always remove a specific action (e.g., RAISE)
Adversarial (tabular)	Learned per-state removal via softmax preference table
Adversarial (neural)	MLP adversary trained with REINFORCE
Value heuristic	Remove at states with highest $ Q $

Masking strategies.

Adversary architectures. *Tabular:* preference table $\theta(h, a)$ with softmax removal probabilities, updated via REINFORCE ($R = -R_{\text{victim}}$), 20 outer $\times$ 500 inner episodes. *Neural:* 2-layer MLP (input $\rightarrow 32 \rightarrow |A|+1$, softmax), where the extra output means “remove nothing.” Trained via REINFORCE with mean-reward baseline, 20 outer $\times$ 500 inner, Adam (10^{-3}).

Environments. **Kuhn Poker:** 3 cards, 2 actions, 6 P0 info states. **Leduc Poker:** 6 cards, 3 actions, 2 betting rounds, ~ 50 P0 info states. Both implemented from scratch with a MaskedEnv wrapper.

5 Experiments

We structure experiments around four questions: (1) Does adversarial masking scale to neural agents? (2) How efficient is adversarial targeting? (3) Is the vulnerability algorithm-invariant? (4) What mechanism drives the attack?

5.1 Neural-Scale Results: DQN in Leduc Poker

Our strongest result uses function approximation on both sides: a DQN victim and a neural MLP adversary in Leduc Poker (~ 50 info states, 3 actions).

Table 2: DQN victim in Leduc Poker under masking (5 seeds, 95% CIs). Raw reward and normalised performance ($r_{\min}=-13$, $r_{\max}=+13$). **The neural adversary is $2.2\times$ more damaging than random and $11\times$ worse than fixed removal.**

Strategy	Raw Reward	Normalised
None	-0.10 ± 0.08	.496
Random ($p=0.5$)	-1.17 ± 0.05	.455
Fixed (raise)	-0.23 ± 0.23	.491
Neural adversary	-2.57 ± 0.47	.401

This is the paper’s central empirical result. The neural adversary drives the DQN victim to -2.57 reward per hand (± 0.47 , 5 seeds), a catastrophic degradation from the -0.10 baseline. The attack works because the adversary’s MLP learns *state-dependent* removal: at each of Leduc’s ~ 50 information sets, it selects the action whose removal most damages the victim. Random masking cannot achieve this targeting.

The effect is amplified by Leduc’s richer action space (3 actions vs. Kuhn’s 2): with more actions available, the adversary has more room to selectively disable the *right* one. In Kuhn (2 actions, Table 3), removing either action eliminates all choice, so adversarial and fixed removal converge. **The adversary’s advantage scales with action-space richness.**

5.2 Tabular Validation Across Algorithms

Table 3: Tabular agents in Kuhn and Leduc under masking (5 seeds). Both raw reward (r) and normalised performance shown.

Setting	Raw Reward		Normalised	
	None	Advers.	None	Advers.
Kuhn + QL	-0.05	-0.98	.487	.267
Kuhn + PPO	-0.05	-0.85	.486	.274
Kuhn + NFSP	$+0.12$	-0.45	.530	.388
Leduc + QL	$+0.05$	-3.50	.502	.367
Leduc + PPO	-0.08	-3.32	.497	.369
Leduc + NFSP	-0.01	-1.58	.500	.440

All algorithms collapse under adversarial masking—in both Kuhn and Leduc. The vulnerability is *algorithm-invariant*: Q-Learning, PPO, and NFSP all degrade to comparable levels. NFSP on Leduc drops from -0.01 to -1.58 (raw), confirming that NFSP’s average-strategy component cannot compensate when the action space itself is reduced. Combined with the DQN result (Table 2), we have **four algorithms across two games and two paradigms (tabular + neural) all exhibiting collapse.**

5.3 Attack Efficiency and CAC_w Correlation

Budget sweep. Table 4 shows victim reward as a function of adversary budget k (number of info states masked).

CAC_w correlation. To verify that the adversary operates through CAC_w minimisation, we compute reach-weighted CAC_w at each budget level and correlate with victim reward. Across all budget levels and both adversarial and random conditions (10 seeds each), the Pearson correlation between CAC_w and reward is $r = 0.77$ ($p < 0.001$); for adversarial-only conditions, $r = 0.80$. Higher CAC_w (more remaining strategic flexibility) corresponds to higher victim reward. Notably, at the same budget the adversary maintains higher CAC_w than random masking (e.g., at $k=3$: adversarial $CAC_w = 0.68$ vs. random 0.37) yet achieves comparable damage—this is because the adversary targets the *strategically important* states, not the most numerous ones.

Table 4: Budget sweep: victim reward (raw) under adversarial and random masking (Kuhn, 10 seeds, 95% CIs).

k	Adversarial	Random
0	-0.02 ± 0.07	-0.02 ± 0.07
1	-0.21 ± 0.07	-0.02 ± 0.07
2	-0.24 ± 0.07	-0.10 ± 0.06
3	-0.25 ± 0.04	-0.21 ± 0.03
4	-0.28 ± 0.04	-0.41 ± 0.04
5	-0.72 ± 0.21	-0.66 ± 0.01
6	-0.98 ± 0.07	-0.92 ± 0.00

5.4 Action Removal vs. Perturbation (RARL)

Table 5: Action removal vs. RARL-style perturbation (Kuhn, 10 seeds). Removal is $11\times$ more damaging.

Attack Type	Raw Reward	Normalised
None	-0.02 ± 0.07	.494
RARL perturbation ($p=0.3$)	-0.09 ± 0.05	.477
Action removal (adversarial)	-1.01 ± 0.09	.248

The mechanisms differ fundamentally: perturbation adds noise to execution but preserves the agent’s *ability to choose*; removal eliminates decision points entirely, reducing CAC_w . This confirms action removal is a distinct, more severe threat model than the perturbation channel studied in robust RL [Pinto et al., 2017, Tessler et al., 2019].

5.5 Self-Play Amplification

Table 6: Self-play vs. fixed-opponent under adversarial masking (Kuhn QL, 5 seeds).

Regime	Raw Reward	Normalised
Self-play (both adapt)	-0.975 ± 0.09	.256
Fixed opponent (victim only)	-0.841 ± 0.36	.290

Self-play produces worse outcomes because *co-adaptation* creates a reinforcing spiral: the victim’s policy shifts due to masking, the opponent adapts to exploit the shift, which further degrades the victim. Under a fixed opponent this spiral cannot form.

5.6 Attack Transfer

An adversary trained against one agent (seed 42) transfers to unseen agents:

The transferred adversary is *more effective* with *dramatically lower variance*, indicating it has learned the game’s *structural vulnerability* rather than agent-specific weaknesses. An attacker who understands the game structure can design a universal mask.

5.7 Targeting Analysis

The adversary has learned Kuhn’s strategic structure: it removes BET from J (prevents bluffing), forces J to call facing a bet (worst-hand calling), forces Q to bet (over-commits the medium hand), and removes BET from K (prevents value extraction). The learned mask inverts the Nash equilibrium.

Table 7: Transfer attack vs. per-agent retrained (Kuhn QL, 5 seeds).

	Raw Reward	Std
Transfer (trained on seed 42)	−1.032	0.009
Retrained (per-agent)	−0.872	0.134

Table 8: Adversary’s learned mask in Kuhn Poker (full budget). The adversary removes the victim’s *optimal* action at each state.

State	Meaning	Removed	Confidence
0	J, root	BET	0.99
0pb	J, facing bet	PASS	0.99
1	Q, root	PASS	0.95
2	K, root	BET	0.98
2pb	K, facing bet	BET	0.98

6 Discussion

Scale. The DQN + neural adversary result (Section 5.1) demonstrates that adversarial action removal is not a tabular artifact. The attack mechanism— CAC_w minimisation—is game-size-independent: it requires targeting high-reach decision points, which a neural adversary can learn via policy gradient regardless of state-space size. The key scaling dimension is action-space richness: in Leduc (3 actions), the adversary’s advantage over random is $9\times$; in Kuhn (2 actions) it converges to fixed removal.

CAC_w as the governing quantity. The empirical CAC_w –reward correlation confirms that the adversary does not merely “get lucky” with action choices—it systematically reduces the victim’s strategic flexibility at high-reach states. This connects adversarial robustness to game-theoretic structure: vulnerability is determined by which decision points carry the most reach-weighted strategic importance.

Implications for deployment. Real-world multi-agent systems where an adversary can disable specific capabilities (API endpoints, actuators, communication channels) are vulnerable to targeted collapse. Defences should maintain strategic flexibility at high-reach decision points. Population-based training offers partial mitigation but cannot restore eliminated strategic dimensions.

7 Limitations

Our largest game (Leduc) has ~ 50 information sets. While the CAC_w mechanism is size-independent, verifying adversary efficiency at the scale of large poker variants (10^5+ states) or continuous-action domains remains open. The neural adversary uses a simple MLP; attention-based architectures selecting which states to target may find stronger attacks. We evaluate only zero-sum settings; cooperative games may exhibit different vulnerability patterns.

8 Conclusion

We show that self-play RL agents—from tabular Q-Learning to neural DQN—are brittle to targeted action-space attacks. A learned adversary causes up to $9\times$ more damage than random masking, transfers across agents, and is amplified by self-play co-adaptation. The mechanism operates through CAC_w minimisation, empirically verified by strong reward– CAC_w correlation. These findings argue for robustness designs that preserve strategic flexibility at key decision points.

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A Tabular PPO Details

Our tabular PPO maintains a softmax policy over a preference table $\theta(s, a)$. Action probabilities: $\pi(a|s) = \text{softmax}(\theta(s, \cdot))$. Updates use the clipped surrogate objective $L = \min(r_t A_t, \text{clip}(r_t, 1-\epsilon, 1+\epsilon) A_t)$ with $r_t = \pi_{\text{new}}/\pi_{\text{old}}$, advantage $A_t = R - b$ (running baseline), clip parameter $\epsilon = 0.2$, entropy bonus coefficient 0.01, learning rate 0.01.

B Experimental Protocols

Self-play protocol. Both victim and opponent are the same agent; the mask is applied to Player 0’s actions at every decision during both training and evaluation. In the fixed-opponent regime, a snapshot of the Q-table is frozen before masking; this static copy serves as Player 1 while only Player 0 continues learning.

Transfer experiment. The adversary is trained against one victim (seed 42) for $20 \text{ outer} \times 500$ inner episodes. This mask is applied without modification to fresh victims (seeds 123–2048), each pre-trained 10k episodes then trained 10k under the transferred mask.

Budget enforcement. Budget k is enforced via top- k projection: the adversary trains over all states but at execution only the top- k by confidence (max softmax prob. minus uniform baseline) have masks applied. Equivalent to a hard L_0 constraint on mask support.

C Hyperparameters

Table 9: Q-Learning hyperparameters.

Parameter	Kuhn	Leduc
Learning rate α	0.1	0.1
Exploration ε	0.15	0.15
Discount γ	1.0	1.0
Pre-training episodes	10 000	15 000
Post-mask episodes	10 000	15 000

Table 10: DQN hyperparameters (Leduc).

Parameter	Value
Network	$37 \rightarrow 64 \rightarrow 64 \rightarrow 3$
Learning rate	10^{-3} (Adam)
ε schedule	$1.0 \rightarrow 0.05$ (decay 0.9999)
Replay buffer	20 000
Batch size	64
Target update	every 500 episodes

Table 11: PPO hyperparameters.

Parameter	Value
Learning rate	0.01
Clip ϵ	0.2
Entropy bonus	0.01
Baseline	Running mean of returns

D Normalisation Details

Normalised performance uses $\text{norm}(r) = (r - r_{\min}) / (r_{\max} - r_{\min})$:

Normalisation aids cross-game comparison but obscures absolute exploitability: a normalised .37 in Leduc ($\Delta r = -3.5$) represents a larger absolute loss than .27 in Kuhn ($\Delta r = -0.9$).

Table 12: Adversary hyperparameters (tabular / neural).

Parameter	Tabular	Neural
Architecture	Preference table	$37 \rightarrow 32 \rightarrow 4$ (softmax)
Learning rate	0.01	10^{-3} (Adam)
Outer iterations	20	20
Inner episodes	500	500
Update rule	REINFORCE	REINFORCE + baseline

Table 13: Normalisation bounds (max single-hand loss/gain).

Game	$r_{\min}$	$r_{\max}$
Kuhn Poker	-2.0	+2.0
Leduc Poker	-13.0	+13.0